UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which tool represents an algorithm using standard graphical symbols?
   A. Compiler
   B. Debugger
   C. Pseudocode
   D. Flowchart

2. What is displayed by the following code?

```
int a[4] = {3, 6, 9, 12};
cout << a[1] + a[3];
```

   A. 15
   B. 9
   C. 18
   D. 12

3. Which operator reads data from cin?
   A. =
   B. %
   C. <<
   D. >>

4. Which item is NOT a programming paradigm?
   A. Object-oriented programming
   B. Structured programming
   C. Procedural programming
   D. Debugging

5. Which of the following are selection structures?
   i. if
   ii. switch
   iii. for
   iv. while
   A. i and iii
   B. i and ii
   C. iii and iv
   D. ii and iv

6. Which statement immediately exits a loop?
   A. return 0
   B. cin
   C. continue
   D. break

7. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. n
   B. p

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C. t
D. i

8. Which part of a for loop is executed after each iteration?
   A. The condition only
   B. The update expression
   C. The loop body twice
   D. The initialisation

9. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. ii and iv
   B. i and ii
   C. i and iii
   D. iii and iv

10. Which keyword prevents a variable's value from being changed after initialisation?
   A. break
   B. const
   C. void
   D. static

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(5 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(8 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 4. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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